SAT MATH

Practice tests • Practice Test 1 • Module 1 (22 questions, 35 minutes)

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Name: _____ Date: _____

**Directions**

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example $3/4$ or $0.75$); give negative answers with a minus sign; do not use mixed numbers.

### Question 1 (Algebra)

If $3x + 8 = 29$, what is the value of $x$?

(A) $7$

(B) $\dfrac{37}{3}$

(C) $21$

(D) $3$

### Question 2 (Algebra)

The function $f$ is defined by $f(x) = 5x - 2$. What is the value of $f(3)$?

(A) $15$

(B) $17$

(C) $13$

(D) $1$

### Question 3 (Data Analysis)

A recipe uses flour and sugar in the ratio $3 : 2$. How many cups of sugar are needed with $12$ cups of flour?

(A) $6$

(B) $8$

(C) $18$

(D) $10$

### Question 4 (Advanced Math)

Which expression is equivalent to $(x + 3)^2$?

(A) $x^2 + 9$

(B) $x^2 + 3x + 9$

(C) $x^2 + 6x + 6$

(D) $x^2 + 6x + 9$

**Question 5** (Algebra) STUDENT-PRODUCED RESPONSE

What is the solution to $2(x-4) = x+5$?

Your answer: _____

**Question 6** (Geometry and Trig)

A rectangle is 12 cm long and 5 cm wide. What is the length, in cm, of its diagonal?

(A) 17
(B) 13
(C) $\sqrt{119}$
(D) 60

**Question 7** (Advanced Math)

What are the solutions of $x^2 - 9x + 20 = 0$?

(A) 4 and 5
(B) $-4$ and $-5$
(C) 2 and 10
(D) $-2$ and $-10$

**Question 8** (Algebra)

What is the slope of the line through $(1, 4)$ and $(3, 10)$?

(A) $\frac{1}{3}$
(B) 2
(C) 6
(D) 3

**Question 9** (Data Analysis)

A jacket costs \$80 before a 15% sales tax. What is the cost including tax?

(A) \$95
(B) \$68
(C) \$92
(D) \$81.20

**Question 10** (Algebra)

If $x + 2y = 11$ and $x - y = 2$, what is the value of $x$?

(A) 5
(B) 3
(C) 7
(D) 4

**Question 11** (Advanced Math) STUDENT-PRODUCED RESPONSE

The function $f$ is defined by $f(x) = x^2 - 3x + 2$. What is the value of $f(-2)$?

Your answer: _____

**Question 12** (Geometry and Trig)

What is the area of a circle with radius 6?

(A) $12\pi$
(B) $36\pi$
(C) $6\pi$
(D) $144\pi$

**Question 13** (Algebra)

Which describes all solutions of $x - 7 \geq 15$?

(A) $x \geq 8$
(B) $x \leq 22$
(C) $x \geq -8$
(D) $x \geq 22$

**Question 14** (Data Analysis)

What is the mean of the data set $4,\ 7,\ 9,\ 12,\ 18$?

(A) 9
(B) 10
(C) 12.5
(D) 14

**Question 15** (Advanced Math)

A town's population is modeled by $P = 2000 \cdot 2^{t/5}$, where $t$ is the number of years after 2020. Which statement is true?

(A) The population doubles every 5 years.
(B) The population increases by 2000 each year.
(C) The population doubles every year.
(D) The population increases by 5% each year.

**Question 16** (Algebra) STUDENT-PRODUCED RESPONSE

For the linear function $g$, $g(0) = 4$ and $g(2) = 10$. What is the value of $g(5)$?

Your answer: _____

**Question 17** (Advanced Math)

What is the solution to $\sqrt{x+7} = 4$?

(A) $-3$
(B) 23
(C) 9
(D) 16

**Question 18** (Geometry and Trig)

In a right triangle, one angle measures $30°$ and the hypotenuse is 10. What is the length of the side opposite the $30°$ angle?

(A) $5\sqrt{3}$
(B) $10\sqrt{3}$
(C) 20
(D) 5

**Question 19** (Advanced Math) STUDENT-PRODUCED RESPONSE

What is the sum of the solutions of $x^2 - 7x + 10 = 0$?

Your answer: _____

**Question 20** (Data Analysis)

What is 45% of 200?

(A) 45
(B) 90
(C) 155
(D) 110

**Question 21** (Algebra)

What is the $y$-intercept of the graph of $5x + 3y = 30$?

(A) $(0, 10)$
(B) $(6, 0)$
(C) $(0, 6)$
(D) $(0, 30)$

**Question 22** (Advanced Math) STUDENT-PRODUCED RESPONSE

If $x^2 = 49$ and $x < 0$, what is the value of $x^3$?

Your answer: _____